GSSS INSTITUTE OF ENGINEERING & TECHNOLOGY FOR WOMEN, MYSURU
Department of Electronics & Communication Engineering | GATE Preparation Library

Comprehensive GATE Notes: Communication Systems

Paper Code: EC • Section 7: Communications (9-12 Marks)

Subject Code: EC-COMM • **Target Exam:** GATE ECE • **Weightage:** 9-12 Marks

1. Analog & Digital Modulation Schemes

- **Carson's Rule for FM:** B_T = 2 (Delta f + f_m) = 2 f_m (1 + beta).
- **Digital BER:** BPSK bit error rate is P_e = Q(sqrt(2 E_b / N_0)). Coherent BFSK has P_e = Q(sqrt(E_b / N_0)) (3 dB worse power efficiency).
- **Shannon Channel Capacity:** C = B * log_2(1 + SNR) bits/sec.

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Carson's FM Bandwidth	**B_T = 2 (Delta f + f_m) = 2 f_m (1 + beta)**	Delta f = Freq Deviation	Captures > 98% of total transmitted FM power
BPSK Bit Error Rate	**P_e = Q(sqrt(2 E_b / N_0))**	E_b = Energy per bit	Benchmark digital modulation for minimum required SNR
Shannon Capacity	**C = B * log_2(1 + SNR)**	B = Bandwidth, SNR = Linear Ratio	Theoretical maximum error-free data transmission rate

GATE PYQ (2 Marks) - AM Power Allocation
Question: A 100 kW carrier is modulated with mu = 0.8 by a sinusoid. Find total power and sideband power.

Step 1: P_t = P_c * [1 + mu^2 / 2] = 100 * [1 + 0.64 / 2] = 100 * 1.32 = **132.0 kW**.
Step 2: P_SB = P_t - P_c = 132 kW - 100 kW = **32.0 kW**.

Essential GATE Exam Pitfalls & Scoring Traps

- **Important:** In Shannon capacity, SNR MUST be in linear units, not in dB.
- **Important:** Matched filter impulse response is h(t) = s(T - t), maximizing output SNR.